

Chapter 7

Re-positioning the human translator as the future expert of GenAI translation in the translation classroom: The results of a collaborative study

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With the launch of ChatGPT in 2022, which allows the production of translation through a prompt-based dialogue between human and generative artificial intelligence (GenAI), translation activity has been reshaped into a genuinely collaborative endeavour. The nature and potential challenges of this emerging form of translation practice call for immediate investigation. To this end, this chapter examines the results of a research experiment in which graduate-level students collaborated with a translator educator to explore the dynamics of prompt engineering as an emerging job role in the field of GenAI translation. The results showed that the experiment helped the students diagnose not only the strengths and weaknesses of GenAI but also their own, acknowledging the need for human intervention, editing and evaluation, to turn raw GenAI output into a functioning translation. The students also re-evaluated their human added value and self-concept as educated and experienced human translators, highlighting the sense of empowerment and responsibility they felt as human decision-makers in the prompt engineering process while addressing areas of self-improvement such as developing skills to communicate with GenAI, which are distinct from those required to communicate with a human being. The results also indicated that the established, holistic skill set of a trained professional human translator was regarded a prerequisite for effective prompt engineering. This needs to be considered by translator educators who now have the task of educating students so that they can cautiously but confidently collaborate with GenAI.



1 Introduction

The impact of machine translation (MT) on the translation profession and translator education has been profound in the last decade, as has the research into this impact. The literature has focused prominently on the tense relationship between the human translator and the machine/technology, which Pickering (2008) conceptualised as a “dance of agency”. Olohan (2011: 354) drew attention to the “decentring” of the human agent in this context. O’Brien (2012: 119–120) further characterised translation as “human-computer interaction”, defining what was happening then as a “shift in paradigm” and pointing out the potential for “dialogue” between human translators and technology.

Ruokonen & Koskinen (2017: 321) further elaborated on the human-machine dynamic, with a focus on the emotional dimension inherent in this relationship. They found that the human agents were both “willing” and “reluctant” in terms of assigning agency to the machine and that the relationship between the human and the machine is too “complex” to be reduced to “simplified man versus machine conceptions”. In a similar vein, Cadwell et al. (2018: 312) observed that human translators held both positive and negative perceptions of MT. Positive perceptions related to reduced effort and increased inspiration while the negative were linked to concerns about the negative influence of MT on human creativity. The authors also argued that the ‘dance of agency’ between the human and the machine was a symbiotic one, as they mutually feed off and shape each other; the “distinction between the human and the material agent” is blurred (Cadwell et al. 2018: 303).

The practical consequences of the pervasive integration of MT technologies into the global translation industry and their economic impact have also been explored. Moorkens (2017), for instance, noted the negative consequences of the freelance employment model, which has been increasingly adopted in the global translation industry, in terms of prices and isolation. According to him, the focus should be on how the human translator actually works with MT. Similarly, Vieira (2018: 15) found that, rather than feeling an “intrinsic negativity to the technology”, the perceived threat from technology among translators appears to stem from business practices in the industry.

Another focus in the related literature has been on the need to redefine the profile of human translators. Sakamoto (2019: 68–69) underlined the necessity of the “redefinition of the concept of ‘translator’, and consequently ‘translation’ *per se*” on the grounds that “the notion of ‘translation’ is now being challenged by the growth of technologies”. The author also noted that attention should be paid

to the uncertainties related to the legal requirements in MT use, pricing for post-editing, and post-editors' profiles and skills (Sakamoto 2019: 63–66). Based on a comprehensive analysis of the emerging needs and requirements in the translation sector, which was conducted with the aim of addressing the “distance between teaching and professional practice” in the context of MT competences, Gaspari et al. (2015: 333–334) reported “a strong need for an improvement in quality assessment methods, tools and training”, underlining the “growing up-take of MT and the perceived increase of its prevalence in future workflows”.

There has been extensive and substantial research into embedding in translator education the new skills, abilities and competencies that human translators should possess in the age of MT, as well as into the concomitant need to re-profile translator education. For instance, Doherty & Kenny (2014: 296–297) designed an SMT syllabus, which foregrounds the “empowerment” of the human translator by enabling him/her to intervene in and add value to the SMT workflows when facing legal, ethical and technical obstacles (Kenny & Doherty 2014: 288). This “empowering” approach has been further extended by Mellinger (2017: 284), who proposes a holistic, “cross-module or cross-curricular integration” of post-editing and MT “to avoid compartmentalisation of various competences and skillsets”. Mellinger also suggests the inclusion of “controlled authoring, terminology management, engine tuning, and post-editing” in translation practice courses, as these are “representative of many of the skills mentioned in recent works on machine translation pedagogy” (Mellinger 2017: 284). Pym (2019) drew attention to the identification of “automation-resistant skills” and their integration into translator education.

The measures to be taken by translator educators to empower students and help them become “aware of their usefulness in order to maximise their agency as translators” has also been the focus in a study by Moorkens (2018: 375–376), who designed a translation evaluation exercise to enable students to “demystify NMT output”. Based on a reconceptualisation of translator competence in the age of MT, Öner Bulut (2019: 3) designed a learning practice to help students “raise their awareness of their professional self-concept as human translators” and suggested the consideration of human translator competence and human translation meta competence. Nitzke et al. (2019: 248) proposed a novel “post-editing competence model”, which included the core competences of risk assessment competence, strategic competence, consulting competence and service competence. Öner & Öner Bulut (2021: 100) investigated translation students' perceived difficulties and benefits in the context of “post-editing oriented neural machine translation error annotation and quality evaluation”.

Exploring the “dynamics of the human-machine dance in the translation classroom”, Öner Bulut & Alimen (2023: 377) designed a learning experiment to enable the students to carry out emergent MT-related tasks of post-editing, pre-editing and error annotation. The authors reported that the experiment helped the majority of participating students “raise awareness of their self-concept as human agents and of the human added value they can create while dancing with the machine” (Öner Bulut & Alimen 2023: 387).

These examples of past studies focusing on the impact of MT on the translation profession and translator education show that translation studies (TS) research has indeed come a long way to come to terms with and adapt to the paradigm shift (O’Brien 2012) induced by the increasingly all-pervasive technologisation of translation, especially since the launch of NMT in 2016.

Yet, the year 2022 presented both translators and translator educators with another, unprecedented challenge. With the launch of ChatGPT in 2022 (OpenAI), which allows for the production of translation through a prompt-based dialogue between human and generative artificial intelligence (GenAI), the early characterisations of translation as human-computer interaction (O’Brien 2012) and dance of agency (Olohan 2011) have become realities in the fullest sense. This poses new questions and opens new areas of research for the TS community, especially for educators and researchers of translator education.

In their above-cited research, Öner Bulut & Alimen (2023: 389) noted that, in the face of the rapid advancements in MT technologies and the uncertainties concerning their integration into education, the re-positioning of translator educators is as necessary as that of human translators. It has also been argued that such re-positioning involves an “awareness of the fact that the translator educator has the primary task of diagnosing and promoting emergent areas of human added value creation in the MT age”, and that this can only be achieved by restructuring the translation classroom as a “platform of collaborative learning, where all involved can work together to discover what remains and will remain human in the MT age” (Öner Bulut & Alimen 2023: 389).

In the face of the paradigm shift induced by the introduction of ChatGPT and other GenAI tools, translator educators should once again work in collaboration with students to explore the dynamics of integrating these tools into their workflow. Motivated by this very need, the present chapter presents the results of a collaborative research experiment designed to explore the dynamics of the human-GenAI dialogue and find the pathways to be followed in incorporating the insights gained into translator education.

The remainder of this chapter is structured as follows. First, the unprecedented practical and theoretical challenges posed by GenAI technology in terms of the

very *raison d'être* of the human translation are explored. Second, information on the design and implementation of the experiment is given. Third, the data obtained from the experiment is analysed. Fourth, a discussion of the results is presented together with conclusions and suggestions for future research.

2 Human added value and the human translator's self-concept in the age of GenAI: Re-visiting the interpretive task and responsibility of the human translator

ChatGPT is a GenAI tool powered by a large language model (LLM) and has not been developed specifically for interlingual translation. However, the revolution it has brought to translation is of great significance as the tool takes prompts from human agents and hence allows them to produce a translation by providing the tool with contextual information and guidance, i.e. by performing prompt engineering (Yamada 2023 and Chapter 5 of this book). This was certainly not the case with MT, which makes GenAI technology a real breakthrough that now requires translator educators to ask and try to answer new questions about human-technology interaction.

In my opinion, in this endeavour we should not attempt to reinvent the wheel. Instead, we should revisit the relevant existing insights into MT, especially those that deal with the integration of MT into translator education, in a way that empowers human translators to create human added value, as outlined in the previous section (see Introduction). This should also involve trying to answer the challenging question posed by GenAI: *what is human in translation?* This is because, unlike MT, GenAI creates the illusion of having an actual dialogue with a non-human agent which pretends to understand, interpret and produce a linguistic message.

This new form of challenge should urge us to revisit the interpretive task and responsibility of the human translator. Accordingly, the approach presented here is performing a backward reading. Rather than giving priority to the investigation of the performance, potential and/or limitations of GenAI, what the present study proposes is the instrumentalisation of the human-GenAI *dialogue* in order to try to reconsider *what is/will remain human* in translation in a new, brighter light and find ways to incorporate it into translator education in the age of GenAI.

In a seminal contribution by Massey & Kiraly (2019) titled "The Future of Translator Education: A Dialogue", Kiraly states that "transcoding (the *mechanical replacement* of linguistic units from a list with corresponding units from a parallel

list) is not at the *heart* of translation at all” (16, emphasis mine) and that, although in the future the translation profession will be “different”, it is “certainly not going to disappear – unless at some point in our evolution we no longer need to *interpret* texts” (16, emphasis mine).

It is no coincidence that Kiraly highlights the act of interpreting as the *raison d'être* of the human translator, which echoes the hermeneutical approach applied to translation by Schleiermacher (1813/1977). According to Hermans (2015: 101–102), Schleiermacher believed that the sole option for a translator is “to act as the hermeneuticist does: to work to attain the best possible understanding of the foreign text which nevertheless remains foreign, and present to the reader [...] exactly that understanding”. Accordingly, the meaning of a text is not encoded in a text to be decoded by the translator, rather it should be interpreted by the translator to attain the best possible understanding, which constitutes the hermeneutical task of the translator. The translator is then required to present this understanding to the reader, which, according to Herman’s reading of Schleiermacher, “exceeds hermeneutics” (Hermans 2015: 99). In a similar vein, Şerban & Larisa (2016: 295) underline that in Schleiermacher’s approach, the meeting between the author and the reader is “mediated and orchestrated by the translator” and the translator assumes “the most active mediating role”.

Although the hermeneutical approach is of critical importance in that it allows the translator room for active mediation, a specific conceptualisation by Arrojo (1997: 18) of the “inevitable interference” of the translator in the context of translation ethics could further the attempt to reconsider *what is human in translation*. Arrojo views translation ethics in terms of the relationship between the “authorial power” of the translator and his/her “ethical responsibility” (Arrojo 1997: 18). Based on a postmodern view of “language and subject”, the author underlines that “[as] no reading can ever aspire to repeat or protect someone else’s text, translators necessarily have the right to exercise their authorial power ‘as long as their game is played up front’” (Berman 1995: 93, cited in Simon 1996: 36). In the view of the author, “[s]uch ‘right’ implies, however, an ethical responsibility which parallels that of ‘original’ authors” (Arrojo 1997: 18).

In the wake of the emerging paradigm brought about by GenAI, this line of thought can be followed by researchers and translator educators to foreground the human translator’s interpretive task and the accompanying responsibility with the aim of exploring human added value (Massey 2021) and human translators’ self-concept (Kiraly 1990, 2000b, Haro-Soler & Kiraly 2019 and Massey & Kiraly 2019) in the age of GenAI.

Drawing careful attention to the human aspect of added value, Massey (2021: 39) characterises human added value as being “manifest in translational decision-

making and problem-solving on a conceptual level that transcends the surface lexical realisations by which meaning is conveyed in source and target texts”. He also observes that “the added value that human translators can and do bring to bear resides in their socio-cultural, socio-technical 4EA (Embodied, Embedded, Enactive, Extended and Affective) cognition, as opposed to disembodied, de-contextualised artificial intelligence” (Massey 2021: 39). In doing so, the author revisits Venuti’s (2019) advocacy of the “hermeneutical model of understanding translation not as ‘the reproduction or transfer of an invariant that is contained in or caused by the source text’ but as ‘an interpretive act that inevitably varies source-text form, meaning, and effect’” and Pym’s (2003) “minimalist definition of translation competence as ‘the ability to generate a series of more than one viable target text (TT1, TT2 ... TTn) for a pertinent source text (ST); the ability to select only one viable TT from this series, quickly and with justified confidence’” (Massey 2021: 51).

The delineation by Massey (2021: 39) of the added value of the human translator vis-à-vis AI strongly captures the conceptualisation of the human translator as an agent who has the hermeneutical task and power of interpreting the text and who bears the responsibility of their interpretive act. This is something that remains quite impossible for the “disembodied, de-contextualised artificial intelligence”. In the age of GenAI, we must therefore pay unprecedented attention to the translator’s “self-concept” (Haro-Soler & Kiraly 2019), which includes “the image of the translator’s social role”, “the translator’s appraisal of his or her competency for translating a particular text” and “an understanding of *responsibility* towards the other actors in the translation context of situation” (261, emphasis mine). Consequently, we believe that the constructs of human added value and of (human) translator’s self-concept should constitute the privileged foci in exploring the human-GenAI *dialogue* and incorporating the gained insights into translator education in a way that helps students raise their self-concepts (Haro-Soler & Kiraly 2019: 261–262) as human translators whose *raison d’être* is to perform the hermeneutical task of interpreting and bearing responsibility. It is with this focus in mind that the following collaborative research experiment was conducted.

3 Design and implementation of the collaborative research experiment

The research experiment, the results of which are reported in the present study, was designed following the tenets of the collaborative research model, imple-

mented by Haro-Soler & Kiraly (2019). It follows the social constructivist and emergentist approach to translator education (Kiraly 2000a: 256), whereby the educator-researcher works collaboratively with translation students “to investigate topics in the domain of Translation Psychology with the goal of having teacher-researchers learn about the translator’s psychological ‘self’ right along with their students”. Accordingly, the research experiment presented here was collaboratively performed by graduate level translation students, who participated in the research on a voluntary basis, and a translator educator (also the author of the present chapter). The aim was to explore the dynamics of human-GenAI interaction as human translators work with GenAI tools as prompt engineers. The student-researcher group consisted of five students enrolled in the PhD programme in translation studies at Yıldız Technical University, Türkiye. All had previous translation education and experience and near-native English proficiency. The educator-researcher was an associate professor of translation studies at the same university.

In the research process, the student-researchers and the educator-researcher collectively decided on the design of the task of prompt engineering in three sessions held online in June 2024. After the design was agreed upon, the student-researchers performed the task individually and reported on their learning/research processes through self-reflexive accounts of their experiences.

The initial decisions made in the recorded online discussions concerned the selection of the source texts and of the GenAI tool to be used to translate the texts through crafting and, when necessary, curating prompts (i.e. prompt engineering; Yamada 2023) so as to pre-process, initiate, produce, edit and evaluate translations that achieved the intended quality level and functions. It was collectively decided to use the free version of OpenAI’s ChatGPT. As for the source texts, two English texts, an essay and a book description, were selected on the grounds that both texts were of mixed forms between expressive and operative text types, while also displaying the features of an informative text type (Reiss 1981: 124–125). Further, the translation of both texts demanded meticulous attention to cultural nuances and context, idiomatic expressions, tone, intent, and linguistic style. The first source text was an essay by Anna Quindlen, published in her “Life in the 30s” column in *The New York Times* (Quindlen 1987). The second source text was the book description of *The Time Regulation Institute*, the English translation of Ahmet Hamdi Tanpınar’s Turkish novel titled *Saatleri Ayarlama Enstitüsü*, translated by Alexander Dawe and Maureen Freely and published in 2014 (Tanpınar 2014).

Next, decisions had to be made concerning the guidelines, namely the two translation briefs and the general principles of prompt engineering. First, the

educator-researcher provided the student-researchers with the existing (albeit limited, due to the fact that it is a relatively new topic) literature on prompt engineering. This led to an exchange of opinions in the first two online sessions. Drawing on those discussions, the educator-researcher then drafted the guidelines for the task and sent them to the student-researchers for evaluation. In the third session, the draft guidelines were discussed and finalised collectively.

The educator-researcher and the student-researchers then had to decide whether to use the finalised translation briefs as the first prompt in the dialogue with the GenAI tool. Taking into consideration the results of the experiments already conducted in other studies on prompt engineering (e.g. He 2024, Peng et al. 2023, Yamada 2023), it was decided not to include the translation briefs in prompt engineering and instead, to use them as guidance for human translators-prompt engineers only. In this way, the human translators were entrusted with the responsibility of the translation act despite performing this act through prompt engineering. The translation briefs for the first and second source texts were as follows:

(1) Translation Brief 1:

“The Name is Mine” is an essay by Anna Quindlen published in her “Life in the 30s” column in *The New York Times* on March 4, 1987. An online magazine website editor in Turkey needs a Turkish translation of the essay and asks you to translate it into Turkish to publish it on the website. The intended audience for the translation is primarily adult readers who would be interested in the works of Anna Quindlen, who writes primarily on feminism and family life and is a Pulitzer Prize-winning columnist, journalist and author. The editor asks you to translate the second and ninth paragraphs of the essay as a test.

(2) Translation Brief 2:

The source text is the book description of the novel *The Time Regulation Institute*, which is the English translation of the prominent Turkish novelist and poet Ahmet Hamdi Tanpınar’s Turkish novel titled *Saatleri Ayarlama Enstitüsü* (1961), translated by Alexander Dawe and Maureen Freely and published by Penguin Classics in 2014. A Turkish literary scholar is asking you to translate the English book description into Turkish in order to examine how the English translation of the book (*The Time Regulation Institute*) is marketed to the English-speaking audience by Penguin Classics.

As for the general principles of prompting, “persona prompt” (He 2024) was selected as prompting method. This meant that prompting began with the statement “You are a professional translator” without specifying any specific domain. It was also decided that prompt engineering would include the following steps: 1) prompting the tool to analyse the source text for the purpose of translation, 2) prompting the tool to translate the source text, 3) prompting the tool to edit its translation through curating prompts where deemed necessary, and 4) prompting the tool to evaluate the quality of, and comment on, its translation. In order to foreground the responsibility of the human translator, a final step, not included within the scope of the prompt engineering task, was added: post-editing the obtained translation output through prompt engineering to produce the translation end-product. Other than these, no specific prompt statements were drafted in order to allow the human translators-prompt engineers some freedom and creativity in their individual dialogue with the GenAI tool. As a result, each source text generated five versions of prompt engineering and five translation end-products.

After completing the task, each student-researcher wrote a report on their prompt engineering experience, the analysis of which is presented below.

4 Analysis of the results: Towards deeper awareness of the power of the human translator vis-à-vis GenAI

After the experiment was completed, the student-researchers wrote retrospective process reports on their experiences working with GenAI by answering the questionnaire drafted by the educator-researcher. Globally, their responses indicated that the collaborative research had allowed them to both evaluate the potential impact of GenAI on the translation profession and re-affirm their role as human agents and experts in translation.

The answers to Question 1 (“What is your general assessment of the experiment?”), indicated that most of the participants came to recognise that human intervention is still needed in several respects. One participant (P4), for instance, noted that during the experiment she “came to the realisation that human translators still have the final say”. Likewise, another participant (P3) stated that despite the “general conception” that LLMs “are already adequate for daily use or even technical translation at some point”, “they would never be able to replace human translators” in terms of “literary texts” and “mixed forms”. Another participant (P1) stated that human translators’ post-editing is “a necessity rather than a preference”, especially for “restructuring expressive language” and “transferring

the style". Participant 2 highlighted her frustration at the AI tool needing "continuous prompting" to edit its translation "until a final translation, which still need[ed] post-editing" was obtained. Another participant (P5) characterised the experiment as being "educating", noting that although she had used "ChatGPT on various occasions", the experiment was the first time that she "actually sat down and tried to generate a translation through prompting".

The answers to Question 2 ("Upon completing the experiment, have you faced any emotional challenges as a human agent collaborating with GenAI to produce a translation? If yes, what were these challenges?") showed that most of the participants faced emotional challenges while working with GenAI. A feeling of frustration was raised by two participants. Participant 1 stated that he got "frustrated" when the LLM did not "directly implement" his suggestions and he needed to "provide prompts again". Another participant (P5) also stated that she "found talking to ChatGPT quite frustrating" and that she realised that she didn't "have the required patience to guide it through everything". One participant (P2) noted that it was "emotionally exhausting to prompt AI for non-literal translation", especially "if a word is used figuratively" as it was "challenging to explain the function of the word and design the prompt so as to reach an acceptable translation". Only one participant (P3) described his feeling as "constant puzzlement" rather than "frustration or anger" as the tool required "an in-depth thought process on how to write the prompts" and that he "felt like the LLM wouldn't be able to complete the tasks" the way he "wanted it to do". The emotional challenge expressed by one specific participant (P4) differed from the challenges experienced by the other participants in that the "feeling of stress" experienced by this participant stemmed from her underestimation of the "depth of AI's analytical capabilities" and its "speed". The participant (P4) also stated she "experienced a lack of confidence while providing prompts" in the translation of the first source text and "became more comfortable" in the translation of the second source text as she familiarised herself with the tool.

The answers to Question 3 ("Upon completing the experiment, what have you discovered about your power vis-à-vis AI and your self-concept as a human translator?") revealed that the participants addressed various aspects of the power of the human translator on the basis of their experience working with GenAI. One participant (P3) described his experience by saying that it felt like being "[a] supervisor trying to teach a very intelligent new employee how to conduct the job and how it can improve itself in certain areas". According to him (P3), the tool can "easily analyse the text" and "find out what kind of strategies could be needed to translate the text for the [target] audience". However, it fails to "produce translation strategies to overcome translation problems" as it remains "too 'faithful'

to the source text” unless the human user “specifically underlines the problems, provides strategies and solutions”. Similarly, Participant 1 emphasised the significance of the “review and intervention” of the human translator for “preserving naturalness and fluency”. According to another participant (P2), her ability “to make a hermeneutical analysis of the texts and guide AI in that respect” fostered her “self-concept as human translator” as she was the one deciding “what kind of a translation is required and when it is acceptable”. The answer by another participant (P4) highlighted the “sense of empowerment and authority” she had while working with the tool in terms of responsibility. This particular participant stated that she had asked the AI bot to take the responsibility for its translation but that it had refused to do so, stating: “[I] strive to provide accurate and high-quality translations based on the text and context provided. However, the final responsibility for assessing and using the translations ultimately rests with the user or the person overseeing the translation project”. Another participant (P5) also underlined how she had felt empowered by saying that “AI is clever and quick but it didn’t go through the life [sic] the way I did”. She also stated that AI “knows everything about the world but has never actually seen it”, which she found obvious in the translation of the essay “The Name is Mine”. It did not show any “sense of true-life experience”.

The answers to Question 4 (“Upon completing the experiment, how would you assess the impact of your previous training and experience on the way you collaborate with GenAI with a focus on your self-efficacy beliefs concerning your ability to prompt and guide GenAI to produce translations of an intended function?) showed that most of the participants established a clear relationship between their previous training and experiences and the way they worked with GenAI. One participant (P3) stated that he believed that his previous experience and training was “vital during the experiment” and that if he had not “received any education in the field” it would be impossible for him “to guide the LLM”. Another participant (P1) stated that “providing the necessary prompt to the AI” requires “the holistic set of skills that constitute human translator competence” that can only be acquired by “translator training”, while another participant (P2) emphasised the significance of the knowledge of “text types and purpose of a translation” in effective prompt engineering. Another participant (P5) stated that thanks to her previous training and experiences, especially in working with and evaluating “automatic translation”, she “was able to spot what does not sound right as the translator and categorise these improvable aspects”. One participant (P4), on the other hand, stated that she had “come to understand that interacting with an AI is distinct from engaging with a human being”. It required skills other than the “social skills” needed to interact with fellow human beings.

The answers to Question 5 (“Upon completing the experiment, do you have a clearer idea of the strengths and weaknesses of GenAI tools in terms of producing translations of intended function?”) revealed that speed was the most appreciated feature of the tool. Yet, despite this, participants highlighted the tool’s “weakness in terms of fluency” (P1) and its “inability to provide adequate translation outputs without feedback from a professional” (P3). One participant (P5) stated that the tool is able to provide “usable output in many ways when guided accurately” but that “it is not completely reliable” given that it is not fully possible to make the tool “achieve what one has in mind”. Another participant (P2) specifically referred to the text analysis ability of LLMs as a strength. Similarly, another still (P4) described the ability of the tool to “deliver rapid and deep analysis” as its “most notable strength”; however she also added, referring to the essay “The Name is Mine”, that the LLM was “unable to convey the feminist perspective” in its translation no matter how hard she tried to prompt the tool in that direction.

The answers to Question 6 (“Upon completing the experiment, how would you assess the contributive potential (if any) of GenAI tools for professional translation workflows?”) revealed that the participants mostly highlighted the time-saving benefits of GenAI. One participant (P1) argued that “refining translations produced by GenAI through post-editing can save time” while another participant (P2) drew attention to the ability of the GenAI tools to yield “faster translation outputs compared to a fully human translation”. Another participant (P3) described GenAI as a “great tool” as it is able to “efficiently analyse a text in detail”. The role GenAI can play in the translation pre-process was also underscored by another participant (P4), who stated that “AI can enhance the first stage in the translation process by offering comprehensive analysis to facilitate translation” and that it can assist the human translator “in the decision-making process by supplying necessary information”. One participant (P5), on the other hand, expressed her reservations. According to her, GenAI “won’t be able to be professionally integrated into workflows until it is tailored to only translation projects”.

The answers to Question 7 (“Upon completing the experiment, how would you define prompt engineering as an emergent job position for the human translator and the skills it requires?”) showed that the participants had gained a clear understanding of the requirements of prompt engineering as an emerging job position. Participant 5 stated that “the understanding of the translation process” is required to “excel in prompt engineering” together with “translation, language and technology competence” and “AI literacy”. Another participant (P3) stated that he believed that many human translators will become prompt engineers as “most LLMs can be customised according to different needs” and that such

people will be necessary for “customising” and “upgrading” LLMs. Prompt engineering was regarded as “an essential and emerging role for human translators” by Participant 4, who also noted the importance of the re-consideration of “the effectiveness of communication skills in the context of prompt engineering” as giving prompts is “directly linked to the communicative abilities” of a human translator. One participant (P1) stated that without the human translator’s skills in “reading, understanding and analysing a text for translation purposes” and the skills in “reformulating a text in the target language in a way that it is appropriate for the purpose”, it would not be possible “to design the necessary prompts to guide the AI effectively”. The significance of prompt design was also highlighted by a fellow participant (P2), who argued that apart from his/her existing skills, the human translator is required to develop his/her skills in prompting through experience.

The answers to Question 8 (“Upon completing the experiment, what factors do you believe should be taken into consideration when integrating GenAI tools into translator education?”) displayed various considerations. Participant 1 highlighted the importance of “instilling in students the awareness that the importance and necessity of the human translator’s added value remain undiminished” as the human translator “retains the role as the expert and bears ultimate responsibility for the target text”. Another participant (P2) highlighted the necessity to “equip translation students with necessary qualifications to make efficient use of GenAI tools” and to encourage them “to test the full potential of GenAI tools to participate in the technology development phase as well”. Another still (P3) focused on the issues of “accuracy” and “bias”, which are frequently associated with LLMs, and stated that translation students “need to be able to critically evaluate AI-generated content and to efficiently conduct fact-checks, cultural appropriations, fitness for scope, etc.” as LLMs “can generate outputs with factual errors, cultural insensitivity, or biases reflecting the data they were trained on”. A similar concern can be found in the answer of another participant (P5), who stated that the bot’s “strengths and weaknesses and limitations (potential biases in training data, the possibility of factual errors, and the inability to capture cultural nuances) should be taken into account” when integrating its use into translator education. Another participant (P4) stated that, based on her experience during this experiment, “it is advisable to incorporate AI into translation-oriented text analysis courses”. She also added that students need to be informed about the “potential disadvantages of relying only on AI, since it basically tends to produce ‘literal’ translations”.

The answers to Question 9 (“Upon completing the experiment, what competences do you believe translation students should have to become experts of

GenAI translation tasks?”) indicated that all participants agreed on the relevance of the already established translator competencies and skillsets for human translators to become experts of GenAI translation tasks. Additionally, one participant (P2) drew attention to the emotional aspect involved in working with GenAI and stated that it is necessary for students to learn to “control their emotions especially in the dialogues with the AI tools” because “by giving agency to the AI tool, the user attributes human features, which makes it even harder to control especially negative emotions”. Likewise, another participant (P4) stated that “the competence to communicate with the AI”, “emotional competence” and “post-editing competence” are fundamental and that “the quality of the translation can be unavoidably influenced when there is a communication gap between a human translator and the AI”. Another participant (P3) stated that students should be familiar with LLMs so that “they comfortably use different LLMs for translation, be aware of LLM-related problems such as LLM hallucination or limitations of each LLM”. According to another participant (P5), “critical-thinking skills” and “creative skills” are needed in addition to “technological competence”, “editing competence”, “cultural competence”, and “research competence”.

The answers to Question 10 (“Upon completing the experiment, what competences do you believe translator educators should have to educate students to become experts of GenAI translation tasks?”) showed that participants saw a parallel between the competences required of students and those required of the educators (P4 and P5) concerning the integration of GenAI into translator education. According to one participant (P1), translator educators need to engage in a “learning process themselves” by “closely following the developments in the field of AI”. Other participants stated that educators should be “users of the state-of-the-art translation technologies” (P2) and should be “able to design courses that combine traditional translation practices with AI-assisted translation workflows” (P3).

Finally, only one participant (P4) provided an answer to Question 11 (“Other comments”), in which she stated that the experiment helped her “identify” her “strengths and weaknesses as a human translator” as well as those of GenAI. She also stated that “communicating and cooperating with the AI” was more “challenging” than she had anticipated and that “upon completing the experiment”, she identified herself “as a *beginner but a quick learner* when it comes to dancing with the AI” (P4, original emphasis).

5 Discussion and conclusion

On the basis of the above analysis of the participants' answers to the questions, it can be claimed that the experiment provided the student-researchers with the opportunity to work with GenAI to produce a functioning target text according to a given translation brief and, in doing so, to evaluate certain aspects of the human-GenAI dance through an experiment-based diagnosis of the strengths and weaknesses of GenAI.

In the accounts of the participants, the main strength associated with GenAI was, rather expectedly, speed. Apart from this, participants placed special emphasis on the text analysis abilities of GenAI, which were considered to be a significant factor that can facilitate and improve the translation process. Notable weaknesses were GenAI's inability to produce a stylistically adequate and fluent translation, its need for continuous prompting from a human translator (i.e., its failure to implement text-specific translation strategies without guidance from a human translator), its potential to generate biased, culturally insensitive and factually inaccurate output which can only be fixed by the careful monitoring and evaluation of a human translator. Last but not least, GenAI's lack of true-life experience was also mentioned.

Further, participants' accounts also highlight that, during the course of the experiment, the student-researchers re-evaluated their human added value and self-concept as trained and experienced translators. Despite their familiarity with, and experience of, the dynamics of human-machine interaction during translation, they recognised the necessity of engaging in a new form of collaboration with GenAI. Indeed, the participants' accounts of their own strengths and weaknesses were closely related to the strengths and weaknesses they had identified for LLMs. The need for a professional human translator to intervene (edit and evaluate) the raw output of GenAI tools to turn it into a functioning translation was emphasised by all participants. This echoes the findings of recent studies on the necessity of a human translator's intervention due to the limitations of GenAI (e.g. Alimen 2023, Öner Bulut & Alimen 2023). Thus, the process by which the students adjusted the LLM output through prompt engineering mirrors the hermeneutic role of the human translator, who interprets the 'meaning' of a text and reframes it appropriately for the target audience. This underscores that GenAI-assisted translation is not merely mechanical but requires deep cultural adjustments and human judgment.

Participants also highlighted the sense of empowerment and authority they felt as decision-makers, who must make decisions concerning translation strategies and prompt the GenAI tool accordingly. This sense of responsibility brought

them power as the ones responsible for the final translation output. This aligns with the ethical and hermeneutic roles of the human translator. However, some of the participants also addressed areas for self-improvement such as developing skills to communicate with GenAI, which is distinct from communicating with a human being, and designing and curating effective prompts to guide AI through the translation process without becoming impatient and stressed. This gives hints about the possible emotional challenges human translators can face while working with GenAI, evidencing the need for “introducing an emotional intelligence dimension into translator training” (Penet & Fernandez-Parra 2023:349). This is something I believe should be addressed in future research on the integration of GenAI into translator education.

The analysis of the results also provided insights into other factors that need to be considered. All student-researchers acknowledged the significance of their previous education and experience in performing the emerging task of prompt engineering. They also agreed that the holistic set of skills of a trained professional human translator is a prerequisite for effective prompt engineering. This had already been suggested in a recent study on the role of human competencies in the “tech-driven language services industry” (Öner & Bengi 2024:92). Yet most of the participants noted that, in addition to this prerequisite, translation students should learn to communicate effectively with GenAI (i.e., prompt it), to critically evaluate and edit its output and to develop an awareness of the fact that human translators are still the ultimate authority with the right to the final say by virtue of the trust and responsibility invested in them. This particular perspective underscores the importance of acknowledging translators’ active role in reconstructing cultural value and meaning within the interactive translation processes facilitated by GenAI. Further, this also points to the significant potential of hermeneutics-based education in fostering a deeper understanding of the human translator’s role in GenAI-assisted translation.

This, in turn, places greater responsibility on translation educators, who, according to the participants, need to become dedicated learners and users of state-of-the-art GenAI technology so that their translation classrooms become a safe space where translation students learn to communicate and dance more confidently with GenAI.

These experience-based insights, which were gained through a collaborative research experiment, shed light on the emerging opportunities, as well as challenges, for repositioning the human translator as the future expert in GenAI translation in the classroom. They also show a possible path to the fulfilment of the translator educator-researcher’s – also emerging – task of re-evaluating

what is/will remain human in translation and incorporating it into education in the age of GenAI.

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